7 What is Coral Worth to Monroe County?

After gaining insight from Monroe County stakeholders on their perceptions of coral reefs and restoration efforts, the Capstone Team took steps to put these topics into context. What benefits do the FRT and restoration provide to the local community? Are the benefits noteworthy? Can these benefits of coral coverage be quantified?

7.1 Coral in the Florida Reef Tract

The FRT provides numerous economic, environmental, and cultural benefits to its bordering communities, especially the residents of Monroe County. Depending on reef type and health (Precht et al., 2022), the FRT provides coastal protection from hurricanes and storm events via wave energy reduction (Ferrario et al., 2014). It also provides a "sense of place" to the humans that visit or reside near the submerged rainforest (Hausmann et al., 2016; Ryfield et al., 2019), and its extensive biodiversity catalog gives the FRT lots of pharmaceutical potential as well. One such example is AZT, an antiretroviral drug discovered in a Caribbean sponge that was revolutionary in treating HIV (Gibson et al., 2008). Marine organisms have high phylogenetic diversity, meaning the likelihood of finding medicinal compounds can be 400 times higher among coral reefs than in terrestrial ecosystems (Bruckner, 2002).

The reef's rapid decline threatens the benefits associated with the FRT. For example, recent global studies estimate that reductions in the structural complexity of reefs, as well as the loss of just 1 m of reef elevation, could result in storms having over twice their current impact on coasts by the end of the century (Beck et al., 2018; Reguero et al., 2021; Yates et al., 2017). A combination of natural and anthropogenic causes is driving this crisis. As a result of warming and acidifying conditions, many reefs have already experienced degradation —such as coral bleaching—and significant changes in species composition and diversity. Latest research projects, with a high degree of certainty, that these changes will continue even if global warming remains below 2°C (Bindoff et al., 2019). Natural stressors, like the ongoing stony coral tissue loss disease outbreak, are also exacerbated by human-driven activities (DCA, 2015; Precht et al., 2016). However, the most severe threats are human driven. Direct anthropogenic threats include bottom-trawling, overfishing, and vessel groundings, while indirect threats include pollution runoff from fertilizers and sewage and the

effects of human-induced climate change such as rising ocean temperatures and ocean acidification (NOAA Fisheries, 2022). These impacts are critical, particularly in the case of the FRT, as Caribbean reefs take longer to recover from environmental stressors than reefs in the Pacific and Indian Oceans (Gibson et al., 2008). Additionally, given its geographic location, the FRT is also vulnerable to hurricane and tropical storm damage, and projections show severe weather events will only increase in frequency and intensity as global temperatures rise (Bindoff et al., 2019).

7.2 Evaluating the Benefits of Coral Coverage

In addition to the environmental benefits the FRT provides, healthy coral reefs contribute to Florida's economy. The economic significance of this ecosystem is recognized so much that in 2018, the Monroe County Board of Commissioners passed a resolution to support H.R.53/S.232 in establishing the Southeast Florida Coral Reef Ecosystem Conservation/Protection Area (Monroe County Commissioners, 2018). In the resolution, the commissioners cite that "from tourism to marine recreation and sport fishing, coral reefs are an important part of the State of Florida's ocean economy and play a central role in the economies of Southeast Florida's communities" (Monroe County Commissioners, 2018). Given the locally recognized connection between Monroe County's economy and its coral reef ecosystem, the Capstone Team sought to understand the estimated financial value of the local coral coverage and its relation to critical industries for the County. This was done by (1) researching economic data on two industries linked to coral health, Tourism and Commercial Fishing, and (2) estimating the square footage of coral coverage within Monroe County (including the Dry Tortugas) boundaries.

Tourism is Essential in Monroe County

Recreational tourism supports between 33% - 75% of the Monroe County economy depending on the methodology used to categorize income. Methods can include income made by residents vs. income earned by workers who do not reside in the County, and the number of services linked to tourism (NOAA, n.d). A study by Rockport Analytics, a global market research and insights firm, analyzed the impacts of tourism on Monroe County across calendar-year 2019 (Rockport Analytics, 2022). The company estimated that of the $2.5 billion of tourism spending in the Florida Keys and Key West, $1.9 billion (76%) remained in Monroe County's local economy. Retainment came through supporting various businesses and jobs and producing state and local government revenue through taxes.

As summarized in the Rockport Analytics report on Monroe County Tourism in, 2019, tourism spending can also be related to employment and tax generation (See Figure 18).

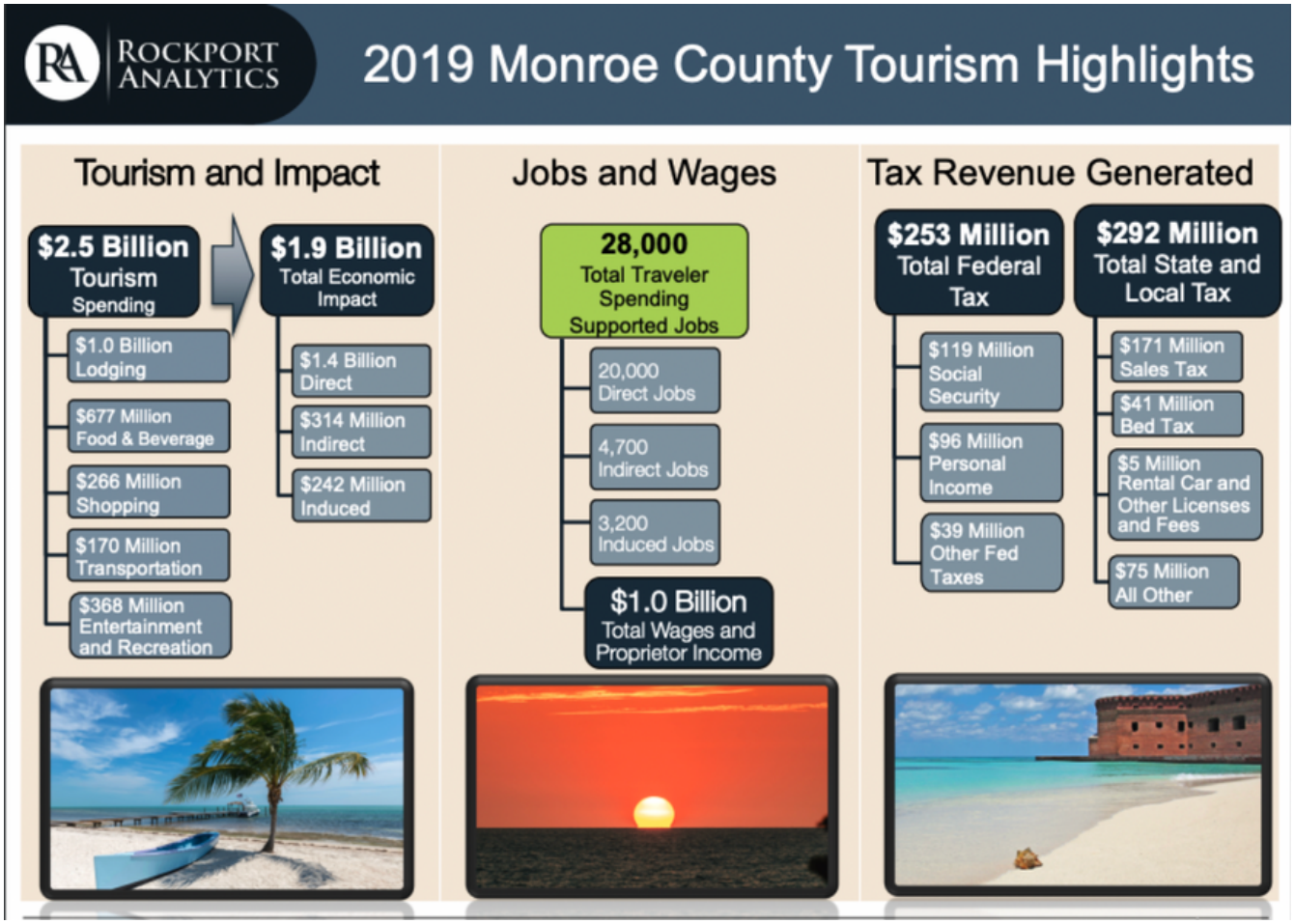

Figure 18. Monroe County Economic Impacts from Tourism in 2019. (Rockport Analytics, 2022)

Employment

Rockport Analytics states tourism in the Florida Keys supported over 28,000 Monroe County residents in 2019 when accounting for direct, indirect, and downstream employment. Tourism represents 43% of all Monroe County jobs, making it the largest employer in the County and a leading economic driver. (Rockport Analytics, 2022)

Tax Revenue

Monroe County tourism generates a substantial amount of state and local tax revenue. Rockport Analytics reported that over $545 million in tax receipts were filed in 2019, of which $253 million (46%) went to the federal government, $160 million (29%) went to the state of Florida, and $132 million (24%) went to Monroe County. In the absence of tourism, Monroe County's ~32,000 county households would need to generate $5,190 in additional local taxes to maintain tax revenues. Tourist-supported state and local tax revenues allow for funding of community benefits such as public schools, relieving household tax burdens, and other public services. (Rockport Analytics, 2022)

Fish Landings Depend on Reef Health

The fishing industry is another clear economic contributor to marine ecosystem health in and around Monroe County. One way to measure the economic value of fishing is based on the estimated value of fish landings. NOAA defines commercial landings as the "weight of, or revenue from, fish that are caught, brought to shore, processed, and sold for profit" (NOAA, 2022). Commercial landings do not include sport, recreation, or sustenance fishing (NOAA, 2022). Based on commercial fisheries landings data reported to the Florida Fish and Wildlife Conservation Commission (FWC), Florida generated an average of $230M/yr worth of commercial fish landings between 2012 and 2021. Monroe County's contribution to this total was about 28% at $63.89 million per year during that period (FWC, 2022).

Of Monroe County's estimated annual landings value, nearly 60% of the total landings value are from spiny lobsters averaging over $37M/yr in dockside value. The yellowtail snapper was the second most valued landing, averaging $6.22M/yr in dockside value. The substantial monetary value attributed to this species is important because once spiny lobsters mature into their juvenile stage, they congregate in protective habitats closer to the shore. Adult spiny lobsters in Florida find protection in crevices and spaces of coral reefs, sponge flats, and other hard-bottomed areas where they forage on snails, clams, and urchins (South Atlantic Fishery Management Council, 2022).To summarize the data evaluated from FWC, Table 5 lists the top nine species that were landed annually from 2012-2021. Even amongst the top 10 species during these 10 years, spiny lobsters led in value by a significant amount.

Table 5. The estimated annual average fish landings value of the top 9 most valuable species in Monroe County. (FWC, 2022)

Species	Average Annual Value (Millions)
LOBSTER, SPINY	$37.14
SNAPPER, YELLOWTAIL	$6.22
CRAB, STONE, LARGE	$4.79
CRAB, STONE, MEDIUM	$3.34
CRAB, STONE, JUMBO	$3.06
SHRIMP, PINK	$2.38
MACKEREL, KING	$0.94
CRAB, STONE, SMALL	$0.86
CRAB, STONE, UNGRD	$0.78

Table 6. The datasets referenced for the Capstone Team's economic research into Monroe County with the time period covered and sources. (Federal Reserve Economic Data, 2022; FWC, 2022; Monroe County Tourist Development Council (MCTDC), 2022; My Travel Intelligence, 2022)

Area	Specific Data Set	Time Period	Source
GDP	GDP for Monroe County	2010 - 2020	Federal Reserve Economic Data
Tourism	Total Leisure Spending Tax Revenue Remitted From Tourism	2015 - 2020 2014 - 2021	My Travel Intelligence MCDTC
Commercial fishing	Commercial fish landings	2012 - 2021	FWC

Data Used for Economic Indicators

For the economic valuation of coral coverage in Monroe County, the Capstone Team researched data in three areas: (1) local Gross Domestic Product (GDP), (2) Tourism, and (3) Commercial Fishing. Based on the datasets, the Capstone Team calculated the annual estimates for 2015-2020, shown in Table 6.

As part of the economic data selection process for the use of the economic valuation of coral coverage in Monroe County, data was chosen based on the following criteria:

- The data is made publicly available by an official source (such as Monroe County or the FWC).
- The data is recent and inclusive of the calendar year 2020.
- The data covers more than three calendar years which allows for adjustment form anomaly years such as 2020 during the COVID-19 pandemic.

Several considerations went into account while reviewing the economic datasets in this report. First, it was challenging to find official and consistent data for tourism. Monroe County's official website links The Rockport Analytics and My Travel Intelligence reports; however, the reported leisure travel spend for 2019 differs. The Capstone Team assumes that the two reporting agencies used different approaches to calculate leisure spending. Unfortunately, neither report included detailed methodologies for the Capstone Team to determine what drives the differences.

Second, the My Travel Intelligence report does not indicate if the annual values follow the calendar or fiscal years. The Capstone Team assumes that the values reported are per calendar year for valuation purposes. Third, the original data on Tax Revenue Remitted from Tourism is based on Monroe County's fiscal year, which runs from October to September.

The Capstone Team converted the values to line up with the January to December calendar year format. Finally, recreational fishing data was not included in the analysis because the available data from FWC is survey-based.

Economic Data Findings

Based on the datasets mentioned in Table 6, the Capstone Team chose a five-year period for the economic valuation. The time frame, 2015-2020, was chosen because data was available for each data set (data was not readily available for each category before 2015). The annual estimates for each economic category and the values used for the economic valuation of coral coverage to Monroe County are shown below in Table 7. From the economic data findings, the Capstone Team observed three insights. First, as expected, economic values were lower during the COVID-19 pandemic year of 2020. Secondly, leisure spending is significant to local GDP. Total leisure spending accounted for more than 62% of local GDP, except for 2020 at 45%. Finally, commercial fish landings make less than 2% of the local GDP; however, this is the dockside value of commercially caught fish. The additional value of fish from other economic activities, such as restaurant and store sales, or out-of-state exports, is not considered.

Table 7. Yearly Estimates for Economic Categories from 2015-2020

Economic Category	2015	2016	2017	2018	2019	2020
GDP for Monroe	$4.43B	$4.69B	$4.93B	$5.04B	$5.47B	$5.26B
Total Leisure Spending	$2.92B	$2.96B	$3.10B	$3.20B	$3.42B	$2.39B
Total tax revenue remitted from tourism	$35.09 M	$36.83 M	$35.66 M	$34.69M	$40.64 M	$31.73 M
Commercial fish landings	$71.26M	$67.68 M	$63.07 M	$70.79M	$58.92 M	$45.62 M

Note. Data sourced from sources listed in Table 6.